

TRACE

Tracking, Recognition, Analytics & City-wide Traffic Enforcement

UI/UX Brief

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Compatibility Note: Compatible with PRD v1.1 and TRD v1.2. No screen, component, or data field specified here was affected by those revisions.

1. Introduction

1.1 Purpose

This brief defines the visual language and screen-by-screen interaction design for the eight screens identified in the Screen Inventory of the App/Website Flow document. It does not introduce new screens, flows, or data — every element specified here exists to serve a functional requirement already defined in the PRD and a flow step already defined in the App/Website Flow document.

1.2 Design Priorities

- Trust over decoration: since every match and alert carries uncertainty, the interface must make confidence visible at a glance, not hide it behind a polished but opaque result (supports NFR-07).
- Map-centric: the GIS map is the primary surface for trajectory and traffic screens, not a secondary widget beside a data table.
- One navigation shell, zero relearning: Operator, Analyst, and Admin see the same shell; only which items are relevant changes (supports NFR-04).
- Calm under alert load: alerts must be noticeable without being alarming enough to distract from an in-progress task.
- Graceful degradation visible in the UI itself: a down camera or low-confidence match is shown as a distinct state, never disguised as normal data (supports NFR-05).

2. Design System Foundations

2.1 Typography

Primary typeface: Aptos, applied consistently across the web application and all project documentation, for a single coherent visual identity between the product and its submission materials.

Role	Weight / Size (approx.)	Usage
Page Title	Bold, 24–28px	Screen-level heading in the content area (not the nav shell).
Section Heading	Bold, 18–20px	Panel/card headings, e.g. "Trajectory", "Segment Detail".
Body Text	Regular, 14–16px	Table cells, descriptions, form labels.
Micro / Meta Text	Regular, 12px	Timestamps, confidence percentages, camera IDs.

2.2 Colour Palette

Colour is used functionally, not decoratively — each colour has one consistent meaning across every screen so the Operator never has to re-interpret it in a new context.

Token	Hex (indicative)	Meaning / Usage
Primary (Ink)	#0F172A	Primary text, headings, nav shell background.
Neutral Surface	#F5F6F8	Page and card background.
Border / Divider	#D9D9D9	Table borders, card outlines.
High Confidence	#1B7A43	Identity Score above threshold; camera "online" status.
Low Confidence / Caution	#B8860B	Identity Score below threshold; degraded camera reliability.
Alert / Anomaly	#B3261E	Blacklist hits, impossible-journey flags, camera-down status.
Heatmap Scale (Low→High)	#2E7D32 → #F9A825 → #C62828	Density gradient on Traffic Analytics heatmap, low to high.

Note: this palette governs the TRACE web application's UI. All project documentation (this brief included) remains black text on white, left-aligned, per the team's document formatting standard — the palette above is content describing the product, not a formatting instruction for this document.

2.3 Core Components

- Confidence Badge — a small pill showing a percentage and colour (High / Low Confidence tokens), used next to every match, plate read, and forecast value.
- Status Dot — green/amber/red dot used for camera online/degraded/down status.
- Alert Card — compact card with type icon, plate, camera, timestamp, and a "Review" action.
- Data Table — consistent striped-row table used for observation lists, blacklist, and OD matrix.
- Map Panel — the shared MapLibre-based component used across Vehicle Trace, Heatmap, and Segment Detail, differing only in overlay layer.
- Empty State Panel — icon, one-line explanation, and a suggested next action; used wherever a screen has no data to show.

3. Navigation Shell Layout

A persistent left-hand vertical navigation bar (collapsible to icon-only on smaller viewports) contains: Dashboard, Vehicle Trace, Traffic Analytics, Alerts, Blacklist Management, Camera Network. The current alert count is shown as a badge on the Alerts nav item, and updates live via the /ws/live-updates channel (App/Website Flow, Flow 3). The top bar carries the screen title, a global plate quick-search field (always accessible, mirroring the Dashboard's quick search), and the signed-in persona indicator.

4. Screen-by-Screen Specifications

4.1 Dashboard (Home)

- Layout: three-zone grid — heatmap snapshot (left, ~60% width), recent alerts list (top right), quick plate search (prominent, top of content area).
- Interaction: clicking the heatmap snapshot navigates to Traffic Analytics — Heatmap; clicking an alert card navigates into Vehicle Trace for that plate (Flow 3).
- States: recent alerts list shows an Empty State Panel ("No alerts in the current window") when empty, never a blank card.

4.2 Vehicle Trace

- Layout: search bar full-width at top; below it, a two-column layout — chronological observation list (left, ~35%) and Map Panel with plotted trajectory (right, ~65%).
- Observation list rows: camera ID, timestamp, thumbnail-style vehicle type/colour tag, and a Confidence Badge; an impossible-journey pair is connected by a red dashed line on the map and both list rows carry an Alert-coloured left border.
- Clicking a row opens an Evidence Panel (right-side slide-over) showing the Identity Score breakdown: plate similarity, OCR confidence, attribute match — each as its own labelled bar (FR-ID-04, NFR-07).
- Time-range filter sits above the observation list as a compact date/time range control.
- Empty state: centered Empty State Panel with "No observations found for this plate in the current data window" and a prompt to check the plate format.

4.3 Traffic Analytics — Heatmap

- Layout: full-bleed Map Panel with heatmap overlay; a compact legend (Heatmap Scale) fixed to the bottom-left corner; a time-window selector fixed top-right of the map.
- A camera/segment with unavailable data is rendered in a flat grey rather than omitted or shown as zero-density green, to avoid misreading "no data" as "no traffic" (NFR-05).

4.4 Traffic Analytics — OD Matrix

- Layout: tabbed within the same Traffic Analytics screen as Heatmap; matrix rendered as a table with origin zones as rows and destination zones as columns, cell shade intensity reflecting trip volume.
- A simple flow-volume bar list beneath the matrix highlights the top five origin-destination pairs for quick reading without requiring the full matrix to be parsed.

4.5 Traffic Analytics — Segment Detail

- Layout: opened by clicking a segment on the Heatmap or OD Matrix tab; appears as a right-side slide-over panel (not a full navigation away), so the Analyst keeps map context.

- Panel contents: segment name, average speed, density, congestion status (Status Dot), and a collapsible Forecast sub-panel showing the short-horizon projection with a visible "heuristic estimate" label (FR-PRD-01).

4.6 Alerts

- Layout: single-column list of Alert Cards, newest first, with a filter bar (type, reviewed/unreviewed) above the list.
- Each Alert Card shows a type icon (Blacklist / Impossible Journey / Duplicate Plate / Camera Inconsistency), plate, camera, timestamp, and a "Review" toggle; reviewed alerts visually recede (reduced-emphasis styling) but remain in the list.
- Clicking a card's plate/body (not the Review toggle) opens Vehicle Trace for that plate, pre-filtered to the relevant time window.

4.7 Blacklist Management

- Layout: watched-plate table (plate, reason, date added, added by) with an "Add Plate" action opening a small modal form (plate number, reason — required fields).
- Newly added plates appear at the top of the table immediately on save, with a brief success confirmation.

4.8 Camera Network Status

- Layout: split view — list of camera nodes with Status Dot and last-seen timestamp (left), map of the same nodes colour-coded by status (right).
- A degraded/down camera row links directly to any trajectories currently showing a gap because of that camera, helping the Operator understand downstream impact rather than viewing the outage in isolation.

5. Map Visualization Guidelines

- Trajectory line: solid line between consecutive normal observations; red dashed segment for an impossible-journey pair (Section 4.2).
- Camera markers: circular markers colour-coded by Status Dot convention (green/amber/red), consistent between Vehicle Trace, Segment Detail, and Camera Network Status.
- Heatmap overlay: continuous colour gradient per the Heatmap Scale token (Section 2.2), refreshed on the live update interval, with a visible legend at all times — never an unlabelled colour overlay.
- Zoom/pan state should persist when a user switches between Heatmap and OD Matrix tabs on the same screen, so context is not lost on every tab change.

6. Responsive Behaviour

The prototype targets desktop/laptop usage as the primary demo environment (an operations-style dashboard), with the navigation shell collapsing to icon-only below a defined breakpoint and two-column layouts (e.g., Vehicle Trace) stacking to a single column with the map above the list on narrower viewports. Full mobile optimization is not required for the prototype demo but the layout should not visibly break on a standard tablet width.

7. Accessibility Considerations

- Confidence and status information is never conveyed by colour alone — every Confidence Badge and Status Dot is paired with a text label or percentage.
- Minimum text contrast follows standard AA guidance against the Neutral Surface background.
- All interactive elements (map markers, table rows, Alert Cards) are reachable and operable via keyboard focus order, not click-only.

8. Traceability

Every screen specified in Section 4 corresponds one-to-one with a row in the App/Website Flow document's Screen Inventory, which in turn traces to specific FR-xxx requirements in the PRD and API endpoints in the TRD. This brief introduces no new data fields — the Backend Schema document defines the exact fields (e.g., Identity Score breakdown, alert type, camera status) that populate the components specified here.

— End of UI/UX Brief —